

Machine Learning Modeling of Earth's Plasma Sheet using Multi-Spacecraft Observations

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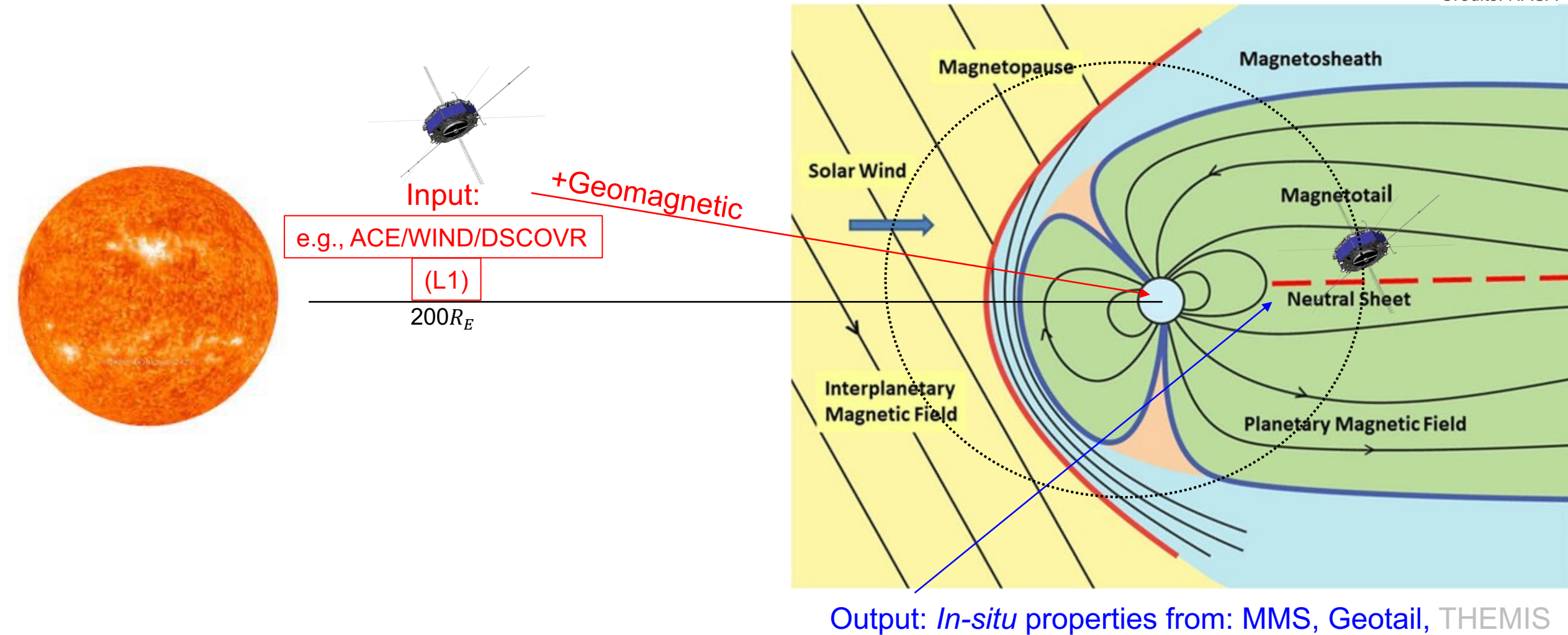
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Acknowledgments:

➤ CGS NASA Drive Center

Where are we & what are we doing?

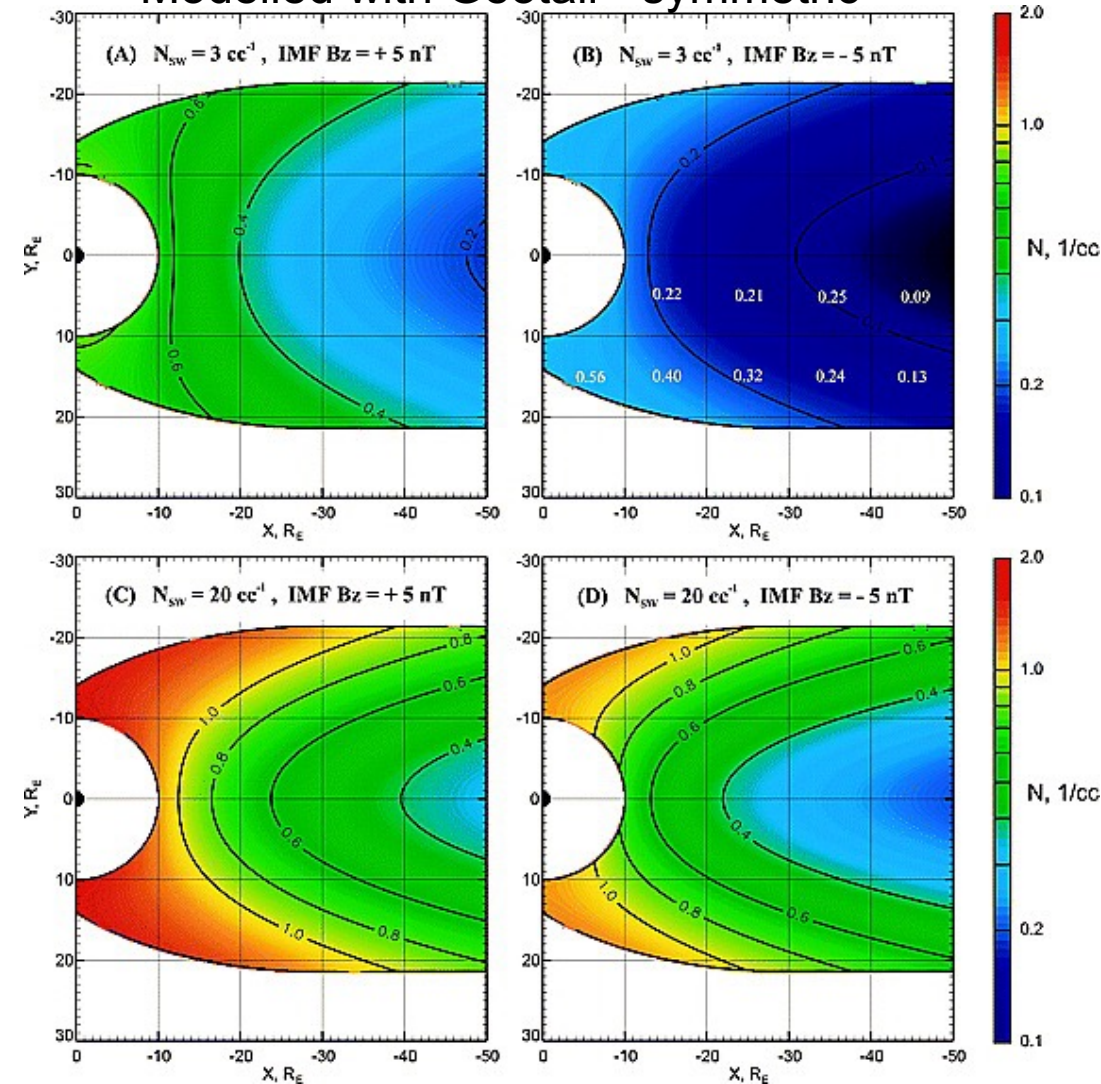
Credits: NASA



Goal: Model Plasmasheet properties based on driving (SW) and geomagnetic conditions

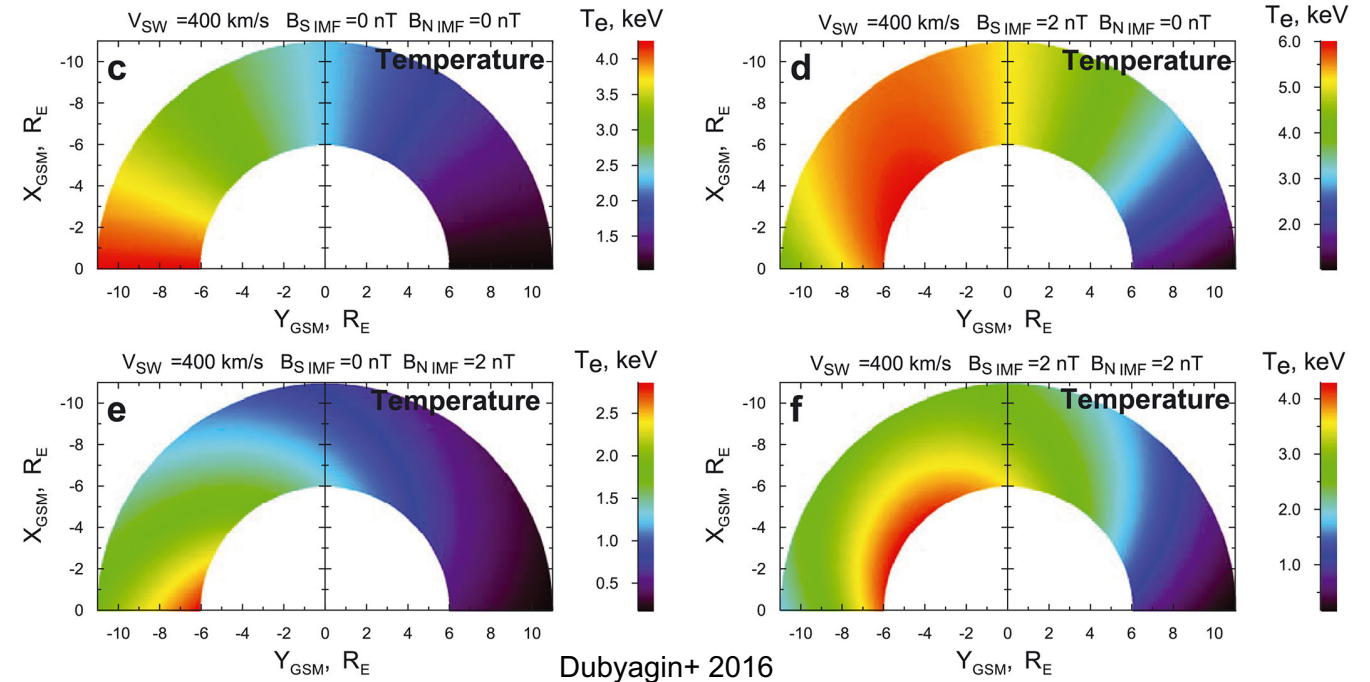
Baseline empirical models

Modelled with Geotail - symmetric



Tsyganenko & Mukai 2003

Modelled with THEMIS



Dubyagin+ 2016

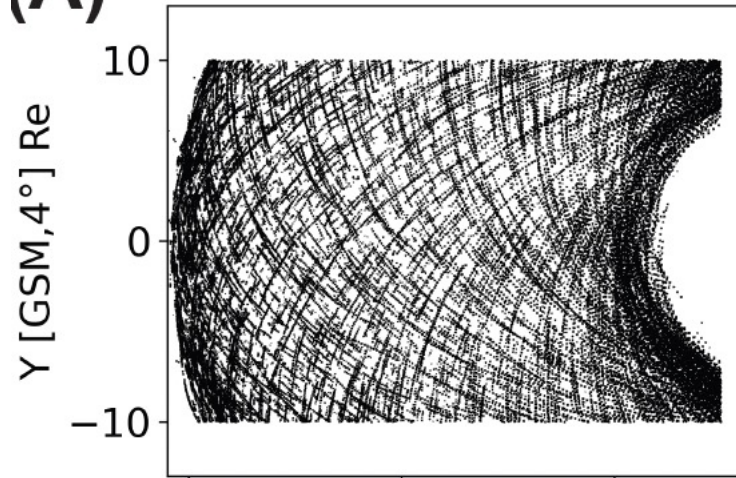
Why then work on this?

1. Far more data, including unexplored MMS instrumentation
2. These models don't include time history (e.g., Borovsky+1998; Wing+2006)
3. Validate with modern and robust statistical/ML practices (i.e., train/test splits)

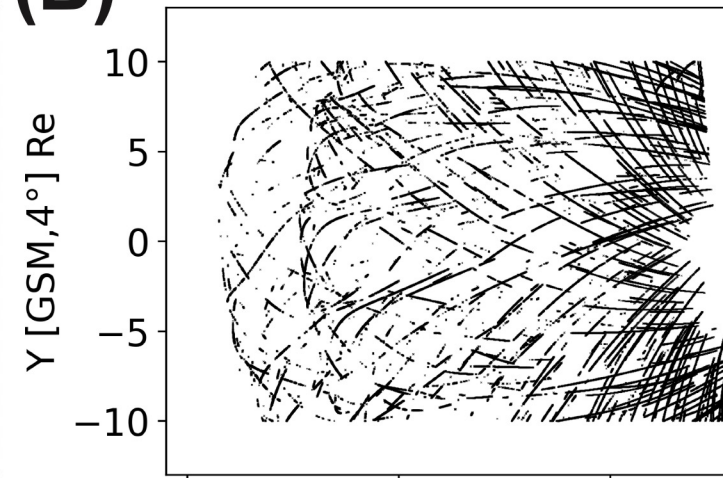
This presentation: Modeling of T_i (MMS), and n_i (Geotail)

The dataset (output – Central Plasma Sheet)

(A)



(B)



(A) Geotail (1994 - 2022)
>1 million points (~12s res)

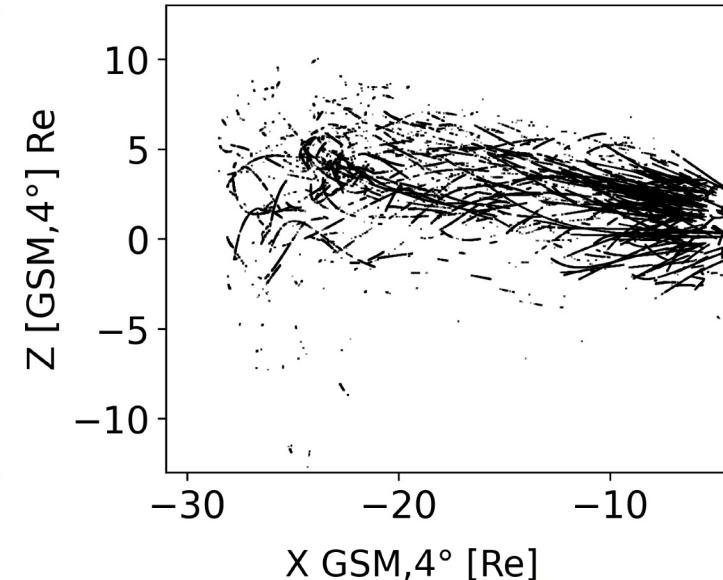
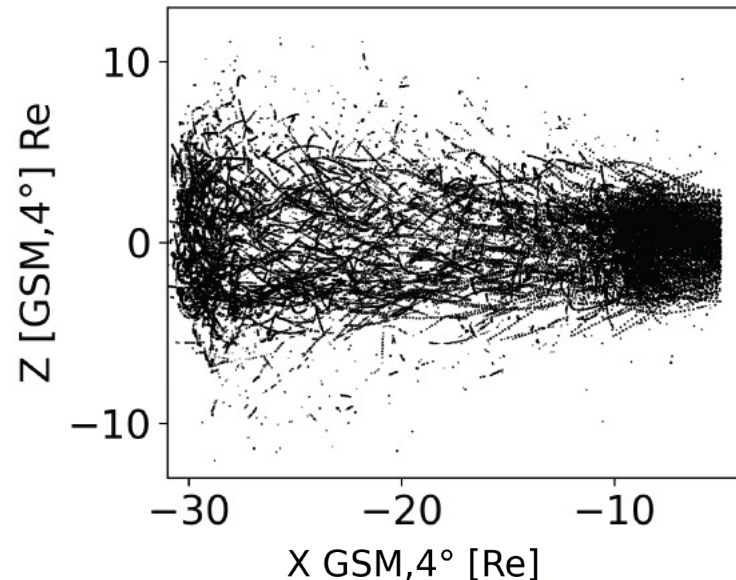
(B) MMS (2015 – 2024)
~ 250k points (~12s res)

Output:

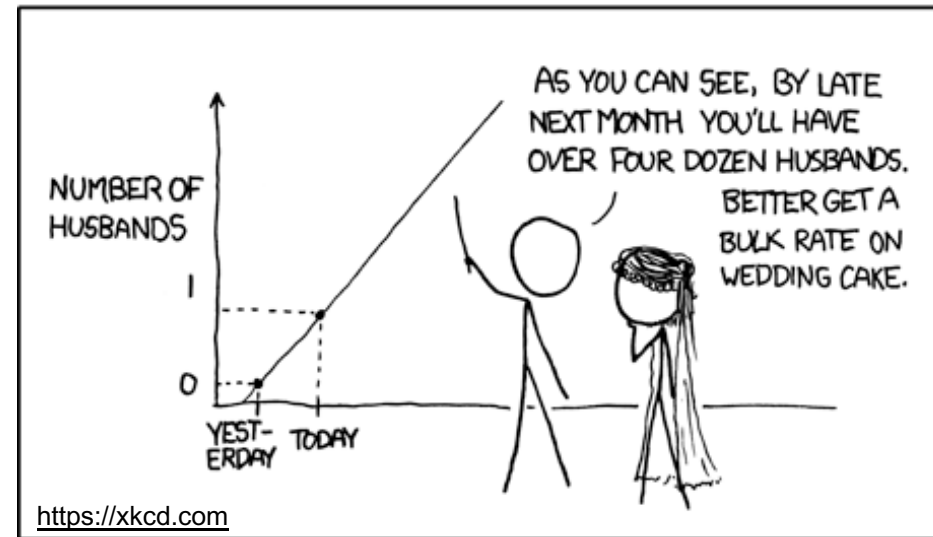
In principle, anything measured
(In this example plasma moments)

MMS moments: partial, not full
Integrated above ~1.5 keV (ions) / ~65 eV (electrons) to remove penetrating radiation and photoelectrons
(*Gershman et al., 2019*).

Shifts mean Ti/Te by ~30%.

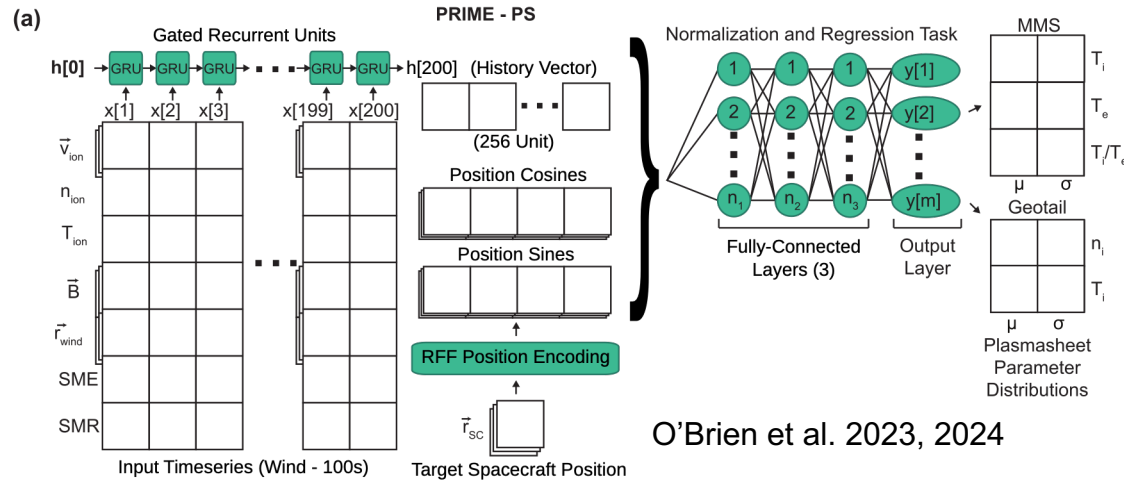


MY HOBBY: EXTRAPOLATING

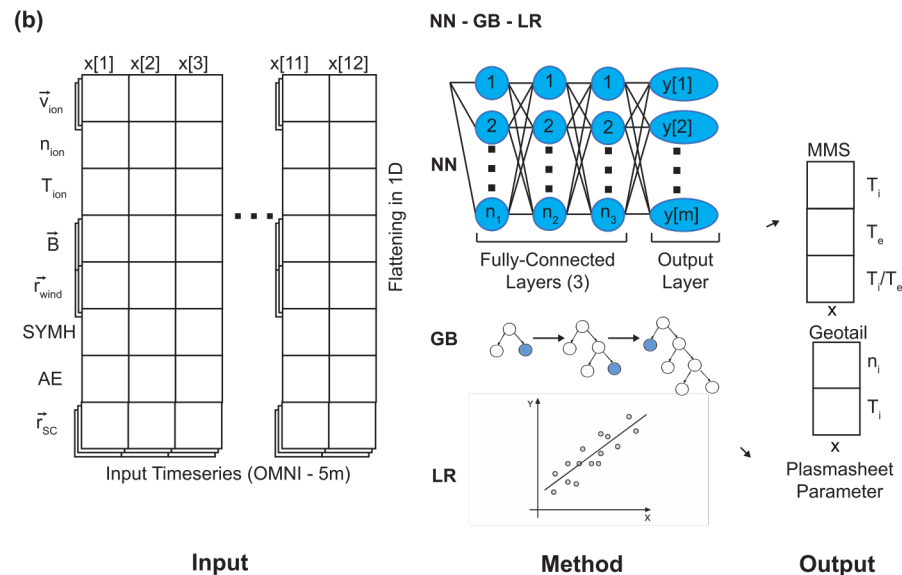


Results

Methodologies & input space



Time History	Type of Input	Architectures
1-6h	Wind (L1)	Linear Reg
	OMNIweb	Gradient Boosting
		Neural Network
		RNN/LSTM/GRU (PRIME-PS)



PRIME-PS: GRU encoder (5.5 h of Wind history) + Fourier-encoded spacecraft position → 3 dense layers → Gaussian (μ , σ) per target, trained on CRPS;

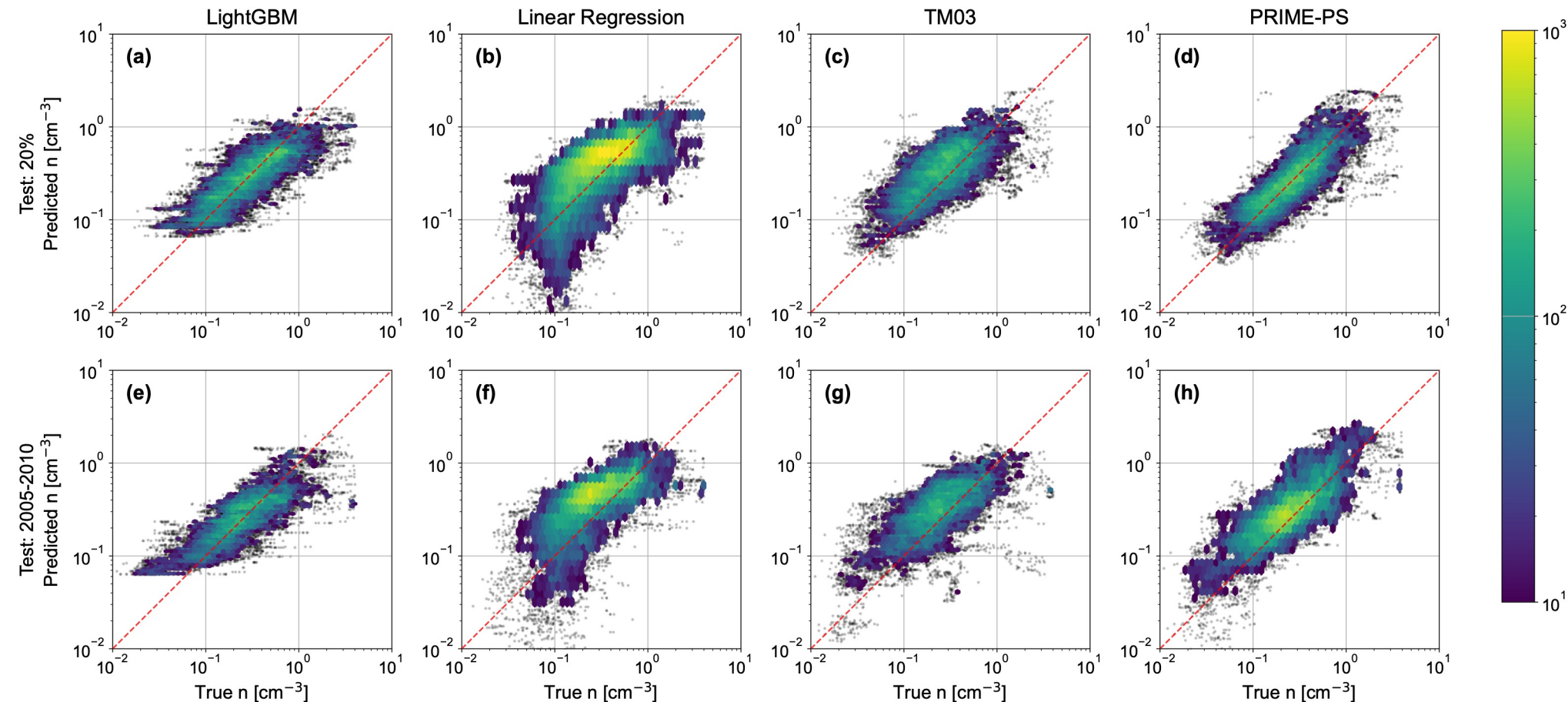
Advantages: Embedded uncertainty quantification and propagation from L1

Key Takeaway:
To quantify our method's impact, we tested multiple variations of the problem.

Modeling Density | Predictions vs Observations

Key Message: PRIME > GB > Baseline $\approx$ TM03

Model Performance | Density (n)



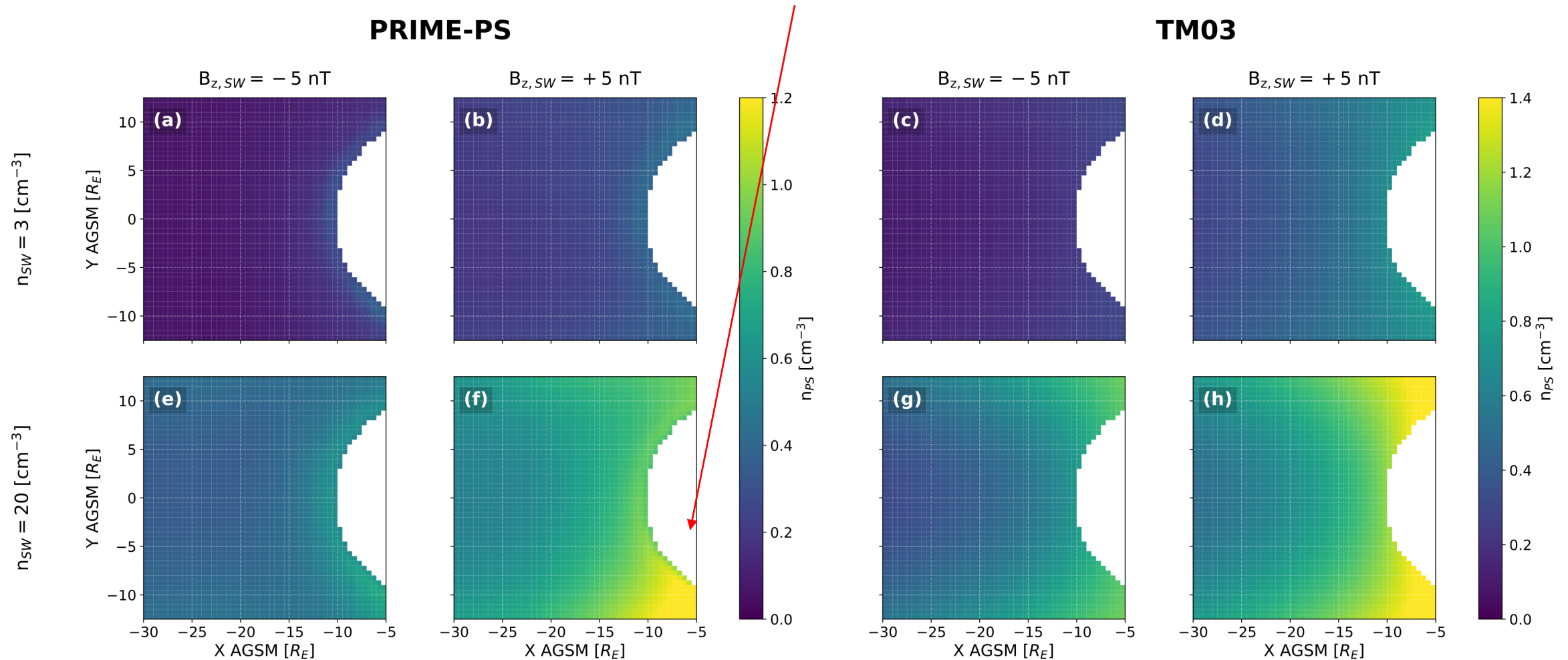
PRIME-PS had a lower MAE by:

~20% vs. LightGBM,
~25% vs. NN,
>30% vs. TM03/linear regression.

✓ Consistent improvements all variables and in other metrics such as Rsquared and correlation.

Modeling Density | 2D Maps (Synthetic input)

Asymmetries Introduced



Neural Networks modeling

$B_z > 0 \rightarrow \uparrow n_{ps}$
 $n_{sw} \rightarrow \uparrow n_{ps}$

Empirical modeling (TM03)

Criterion	Strict CPS	Flexible CPS	High density
$\beta > 1$	yes	—	—
$\beta > 0.5$	—	yes	—
$\sqrt{B_x^2 + B_y^2} < 2 B_z $	yes	—	—
$N < 6$	yes	—	—
$N > 6$	—	—	yes
$EA1SW0 = EA$	yes	yes	yes
$-31 < R_x < -5$	yes	yes	yes
$ R_y < 15$	yes	yes	yes
$ R_z < 10$	yes	yes	—
$V_x > -20$	—	—	yes

Table 1. Plasma sheet classification thresholds for the strict CPS, flexible CPS, and high-density subsets. *beta* is the ion plasma beta parameter, density (*N*) is in 1/cc units, *V_x* is in km/s, and all the locations (*R_{x,y,z}*) are in Earth radius. The coordinate system for all vectors is the aberrated Geocentric Solar Magnetospheric (GSM) coordinates

Storm Time & Extreme Events

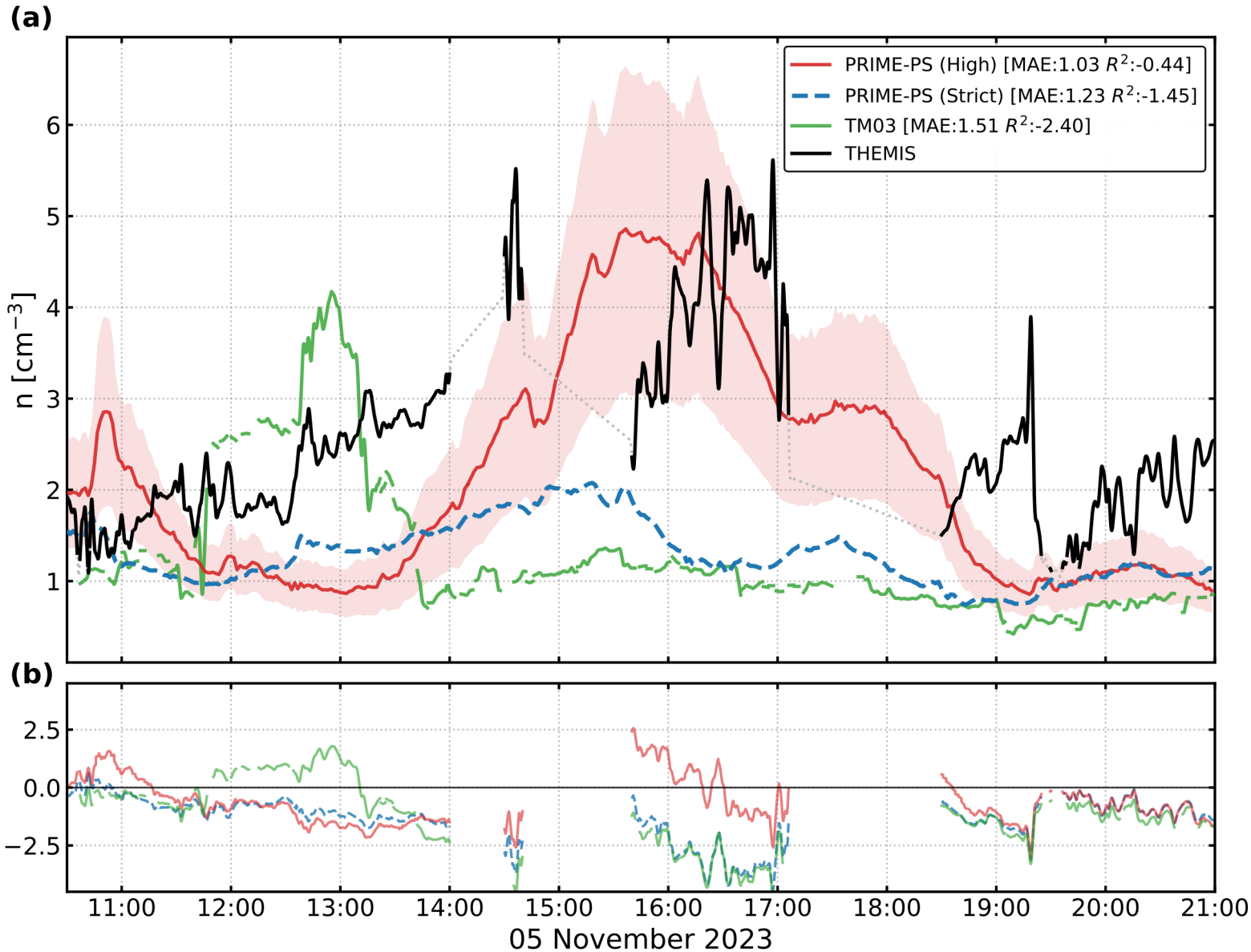
The Problem: We use static thresholds for dynamic environments.

The Risk: We can mistakenly remove crucial "stormtime plasmasheet" datapoints.

The "Fix": Manually find the missing data and add it to the dataset.

Strict CPS (e.g., Ohtani et al., 2008, Raptis et al., 2024) & Flexible CPS (e.g., Richard et al., 2022)

Test case of a storm (05 Nov 2023) - Density



Key Message: >30% improvement over TM03 during a storm

Process:

High: Includes manually picked high-density intervals from Geotail

Strict: Normal threshold-based classification of plasma sheet

Test:

THEMIS observations

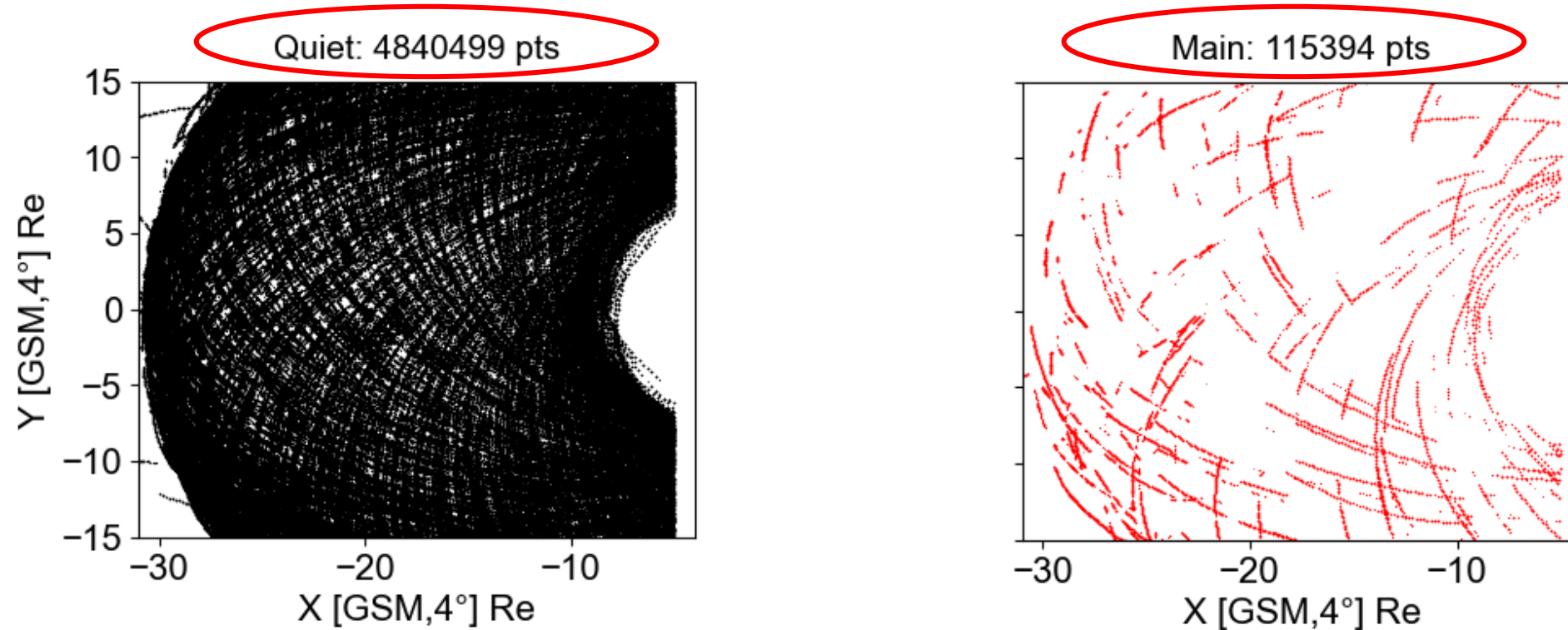
Results

- ML model (high dens): 1.03 [1/cc]
- ML model (normal): 1.23 [1/cc]
- TM03: 1.51 [1/cc]

Note: values $<1 \text{ cm}^{-3}$ are boundary layer transitions, gray shaded linear interpolation

Community Reminder: Data Sparsity During Extreme Events

Geotail data (1994 – 2022), time resolution: 12 seconds



(#of unique days, # unique storms)

Geotail						
Main	7377 (36 32)	3006 (22 19)	2715 (17 17)	6240 (22 19)	6009 (21 19)	2450 (24 23)

Keypoint: Number of data points can be misleading

Note: This doesn't show the incomplete driver (SW) distribution.

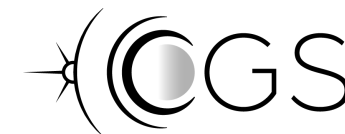
Summary & Discussion

Results

- **Data-driven models reproduce dawn-dusk density, temperature, and Ti/Te asymmetries** that previous analytical models miss
- **Dataset curation**, namely plasma sheet classification and moment contamination correction, critically **controls model performance**
- **Training on high-density storm intervals reduces prediction error by ~25%**, suggesting data sparsity is a primary bottleneck

Discussion Points/Opinions

- Reminder GIGO applies to all data driven models https://en.wikipedia.org/wiki/Garbage_in,_garbage_out. The difference between partial vs. full moments for MMS alone can shift Ti/Te by ~30%
- See our latest EOS article on Heliophysics and AI discussing data sparsity: <https://eos.org/science-updates/vast-space-sparse-data-an-ai-answer-to-twin-space-weather-challenges>



CENTER FOR
GEOSPACE STORMS

CGS Graduate Fellows

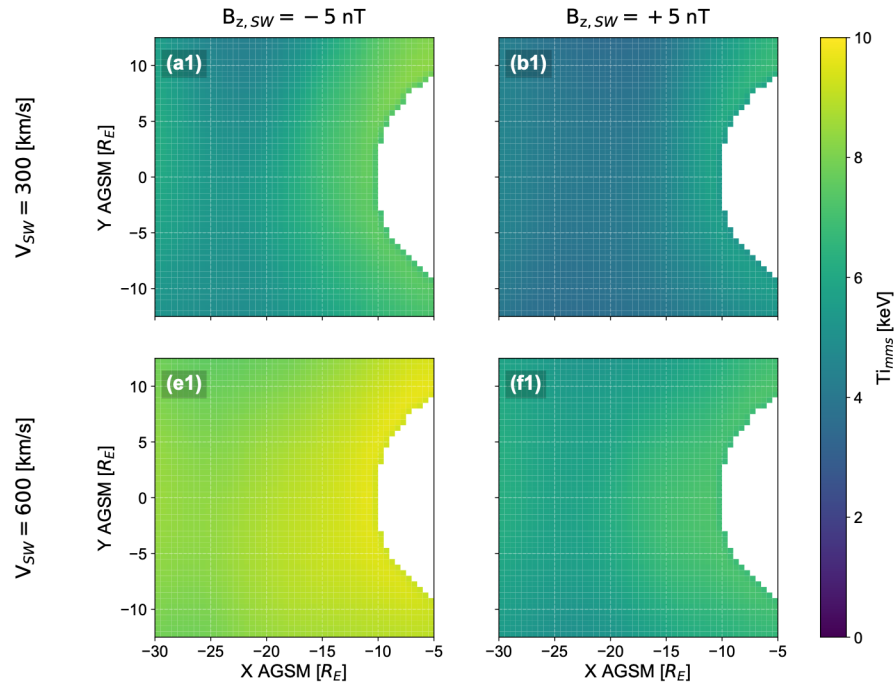
- CGS will select up to 9 CGS Graduate Fellows who will receive support to enhance their current research projects
- Eligibility: graduate students who are US Citizens
- A list of CGS Team Members and their preferred topics will be advertised on the CGS website (will be available by August); Interested students must contact a CGS Team Member to discuss potential projects and level of support provided
- Fellows will receive a stipend (~\$10K-\$12K) to be used for
 - Travel to GEM or CEDAR in summer 2027 to present results
 - Travel to CGS Workshop in August 2027
 - Other expenses at the fellow's discretion
- Applications (due early October) will include:
 - Form with general information
 - 2-page proposal describing project and how it relates to and enhances your current research
 - 1-page letter from research advisor recommending student and approval of project



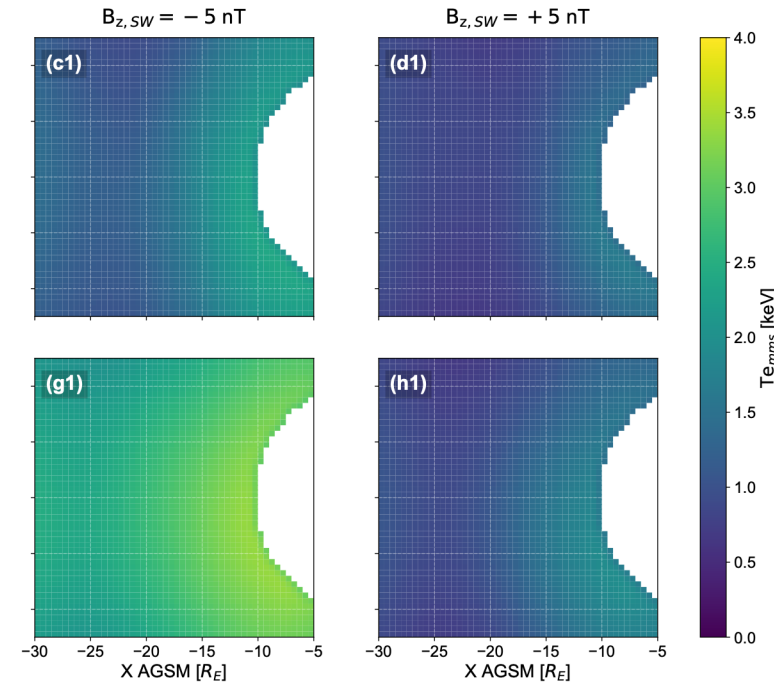
Extras

Modeling Temperatures with MMS | 2D Maps

PRIME-PS MMS Ti

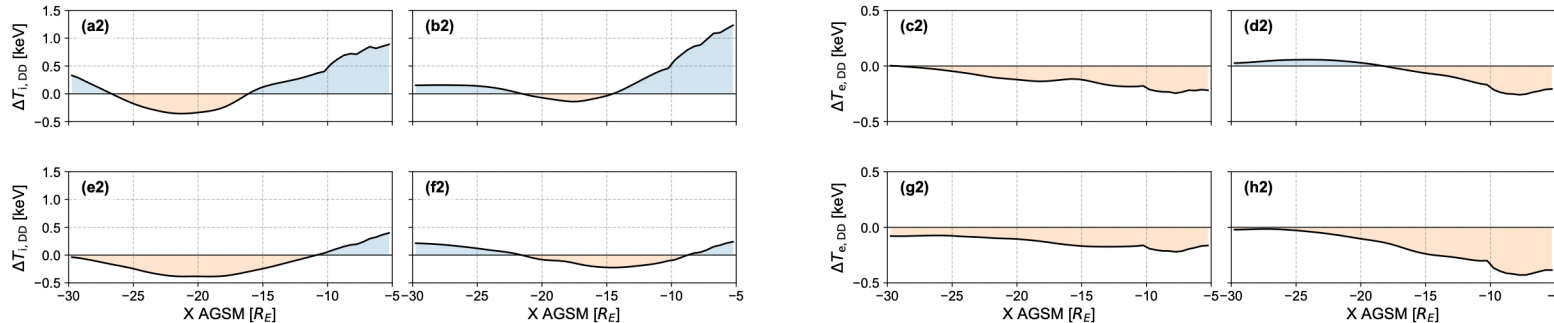


PRIME-PS MMS Te



- Ti: increases with Vsw and peaks **duskward**
- Te: weaker, opposite (**dawnside**) enhancement
- Consistent with charge-dependent gradient-curvature drifts
- **Caveat: dawnside Ti feature (X $\approx$ -25 to -15 RE) may be a sampling artifact; the Geotail-trained model shows none**

Dusk-dawn asymmetry: $\Delta T_{DD}(X) = \langle T \rangle_{0 < Y < 15[R_E]} - \langle T \rangle_{-15 < Y < -0[R_E]}$

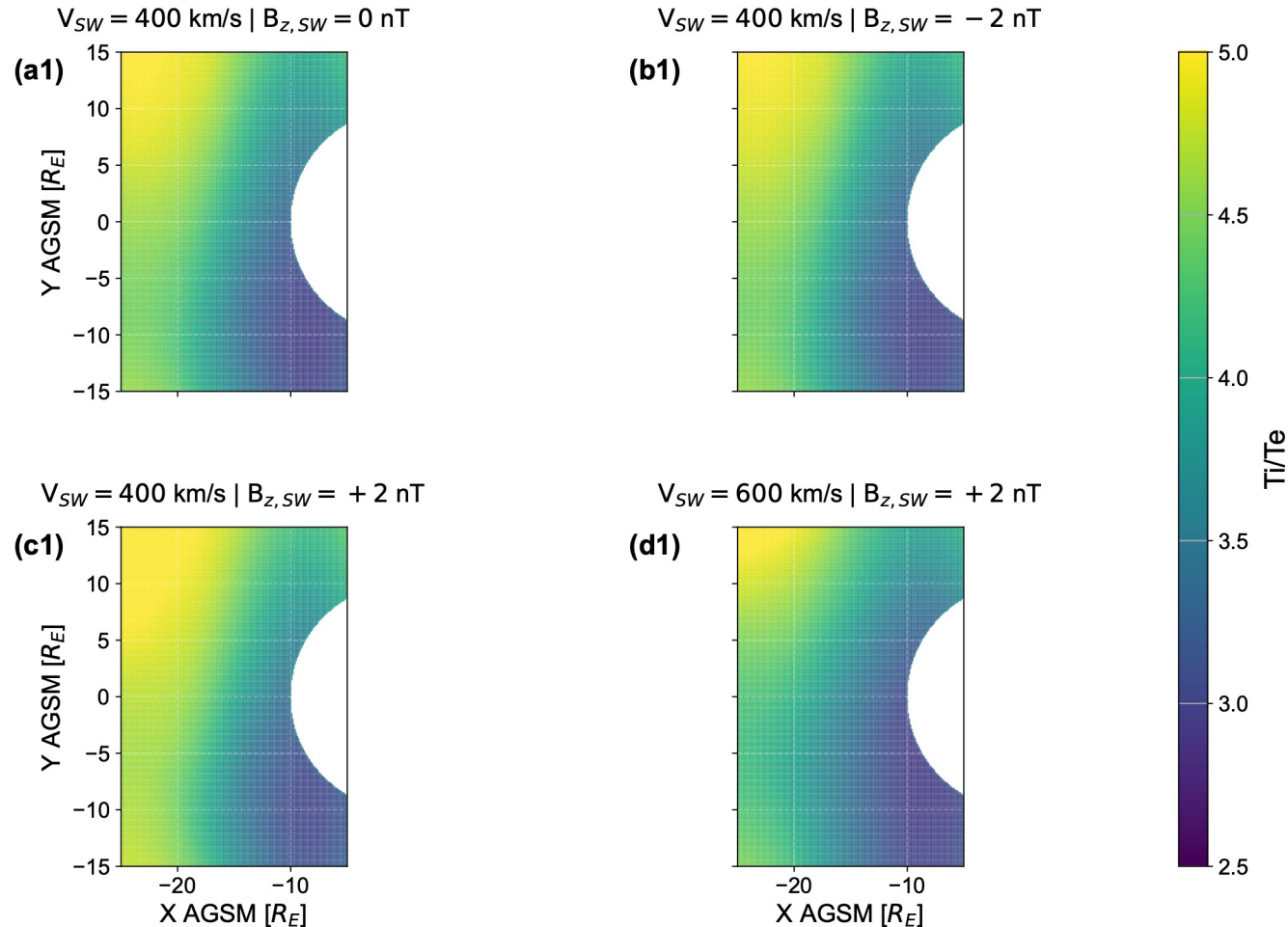


Blue: more dusk, Orange: more dawn

Modeling Temperature Ratios with MMS | 2D Maps

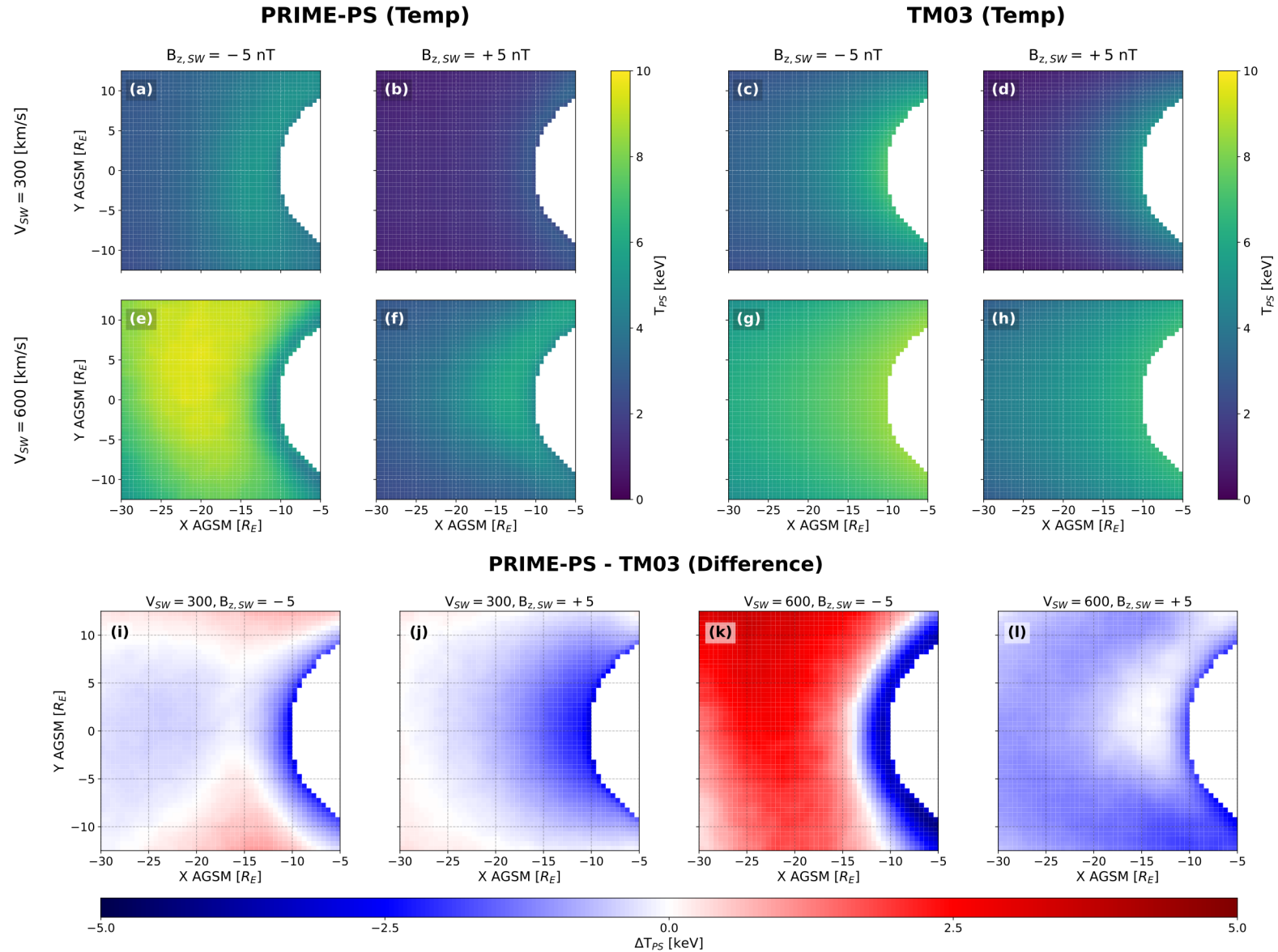
Note: Agreement Wang et al., 2010 with dusk Ti/Te higher than dawn (Using THEMIS)

PRIME-PS (Ti/Te)

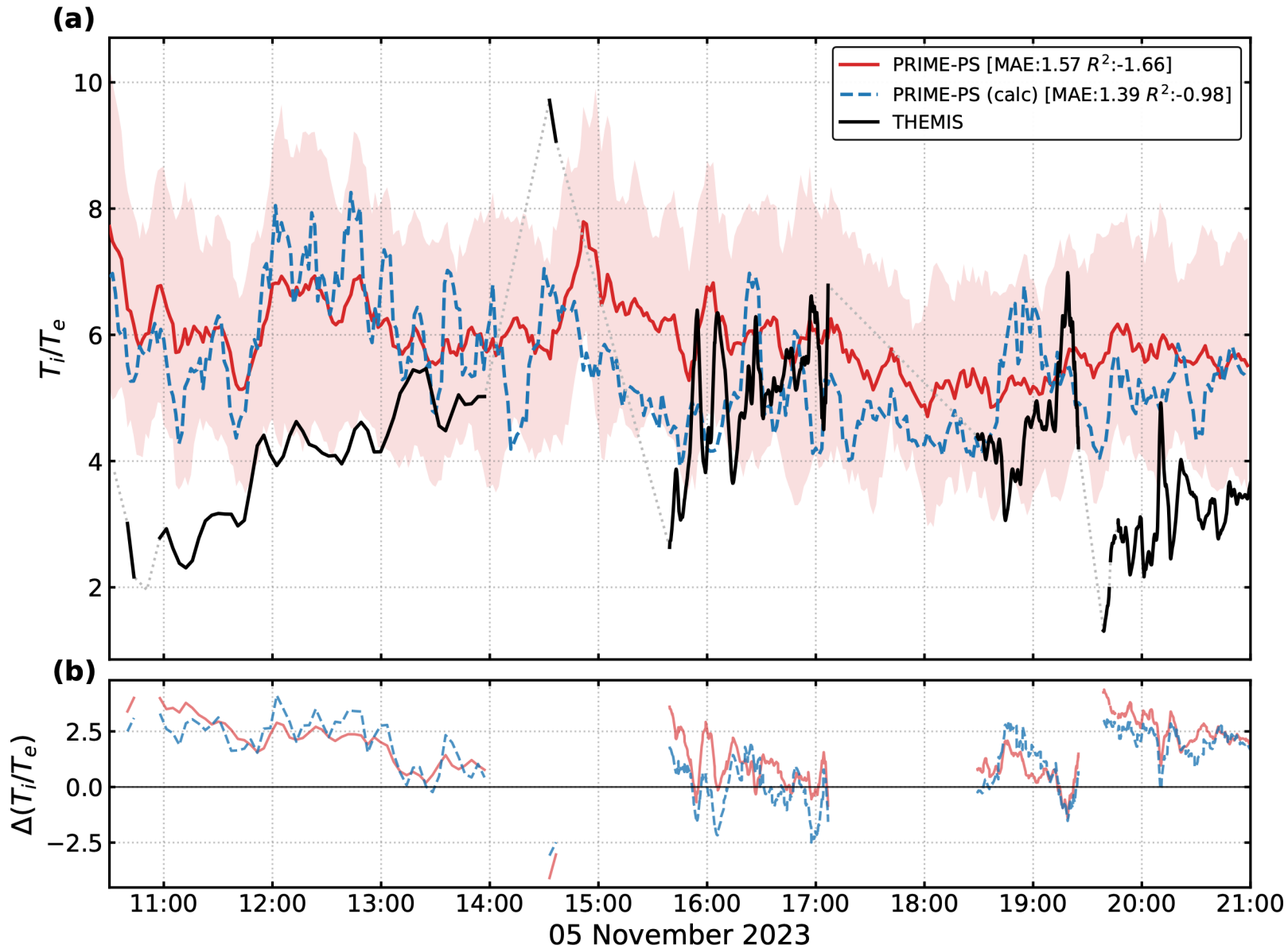


- Ti/Te rises from ~3 on the dawn side to ~5 on the dusk side
- Duskward enhancement slightly stronger under northward IMF
- **Caveat:** for northward IMF the asymmetry weakens as V_{sw} goes from 400 to 600 km/s, but fast-wind northward intervals are scarce in training, so this may reflect uneven sampling
- Analytical alternative (TM03 ions + DGSR16 electrons) mixes spacecraft with non-overlapping domains and produces extreme values; ML gives a self-consistent ratio

Modeling Ion Temperature with Geotail | 2D Maps

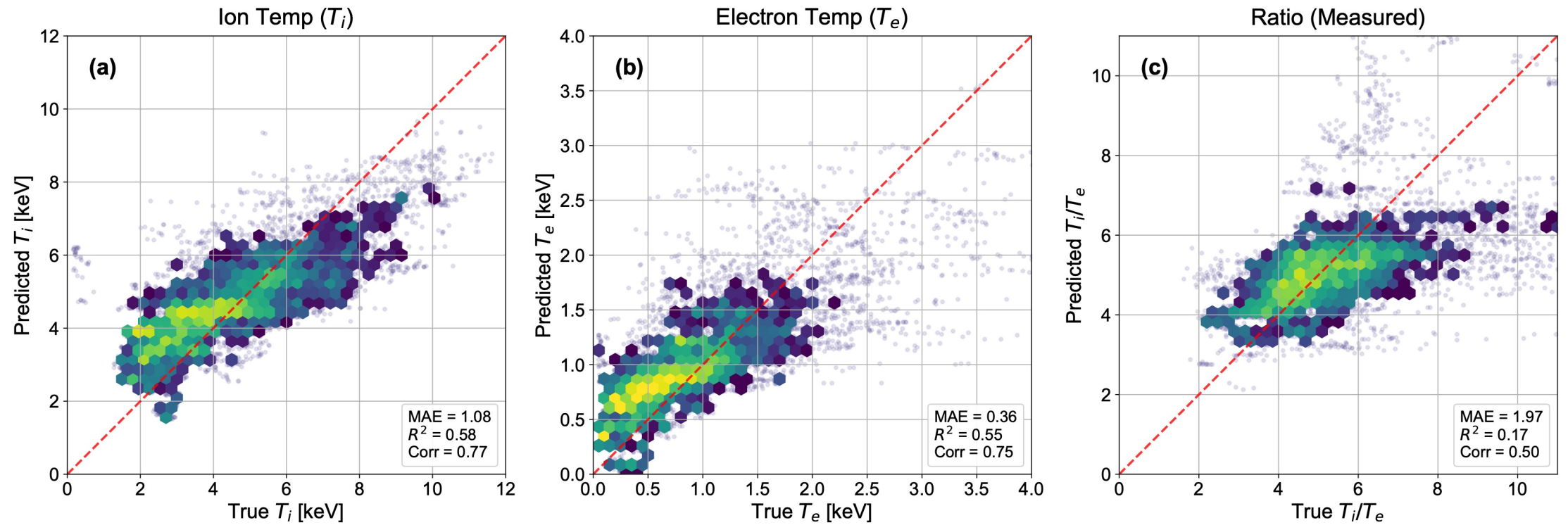


Test case of a storm (05 Nov 2023) – Ti/Te



Note: values $<1 \text{ cm}^{-3}$, are boundary layer transitions, gray shaded linear interpolation

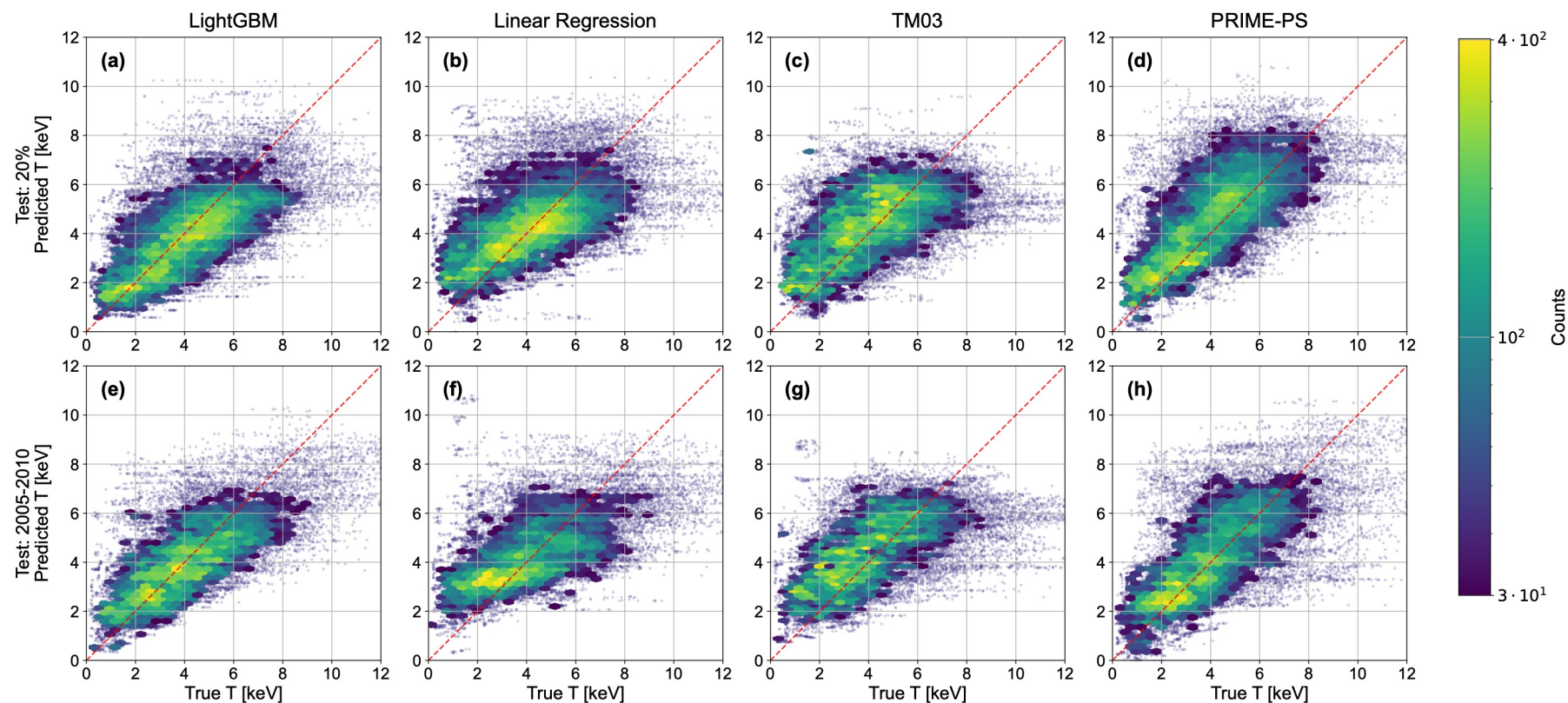
Temperature and Ti/Te MMS output



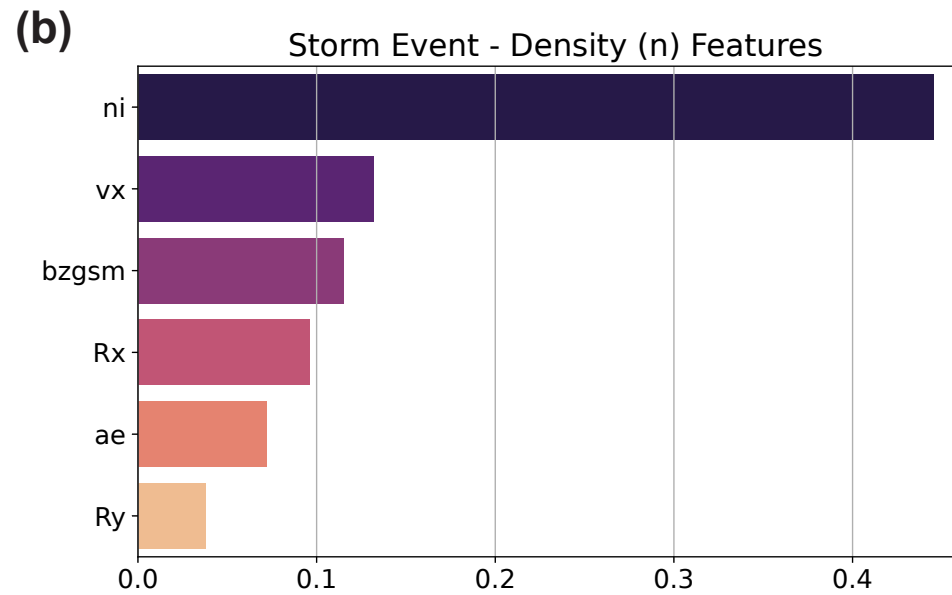
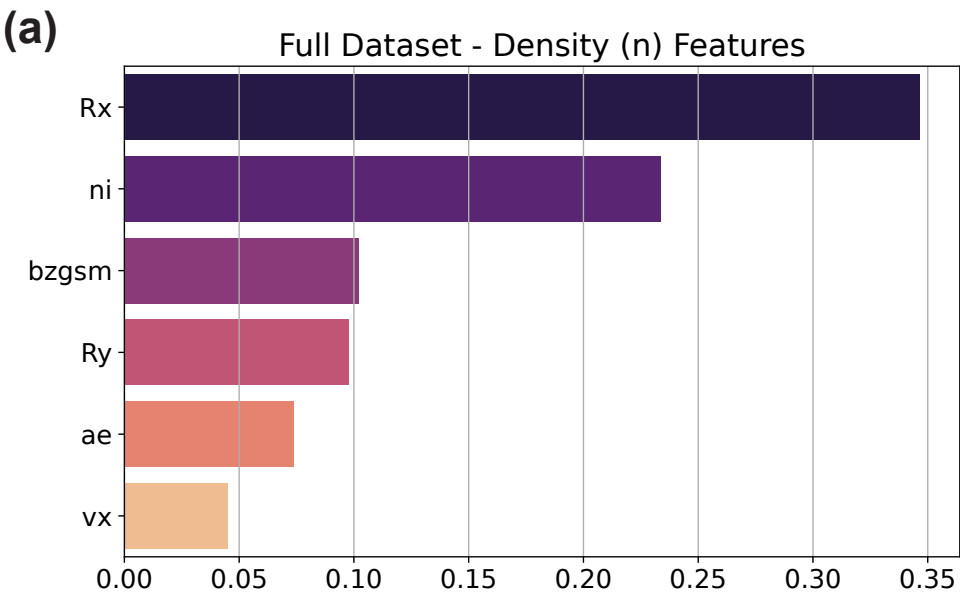
Modeling Temperature | Predictions vs Observations

Key Message: PRIME – PS > GB > Baseline $\geq$ TM03

Model Performance | Temperature (T)



Feature Importance Analysis – Merged Time history

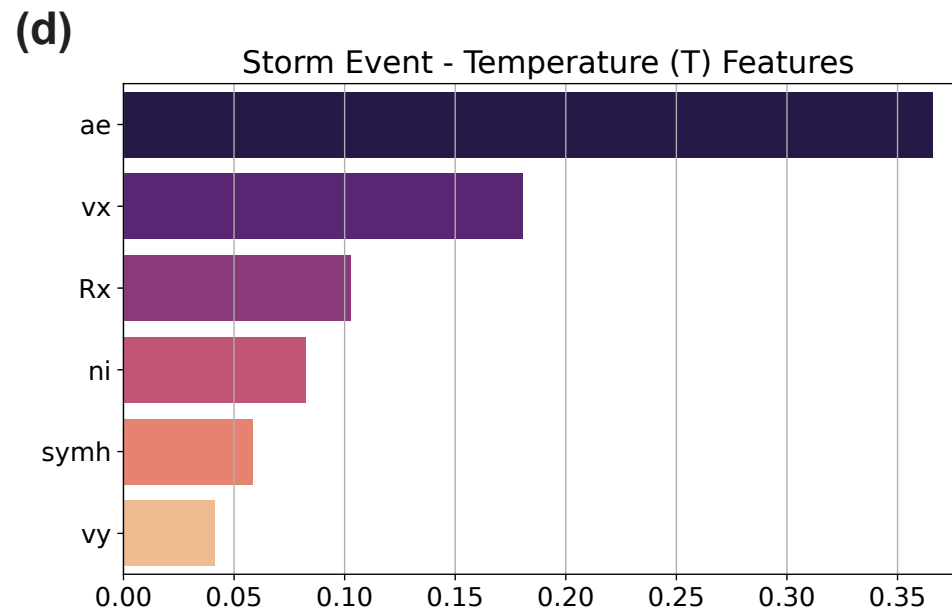
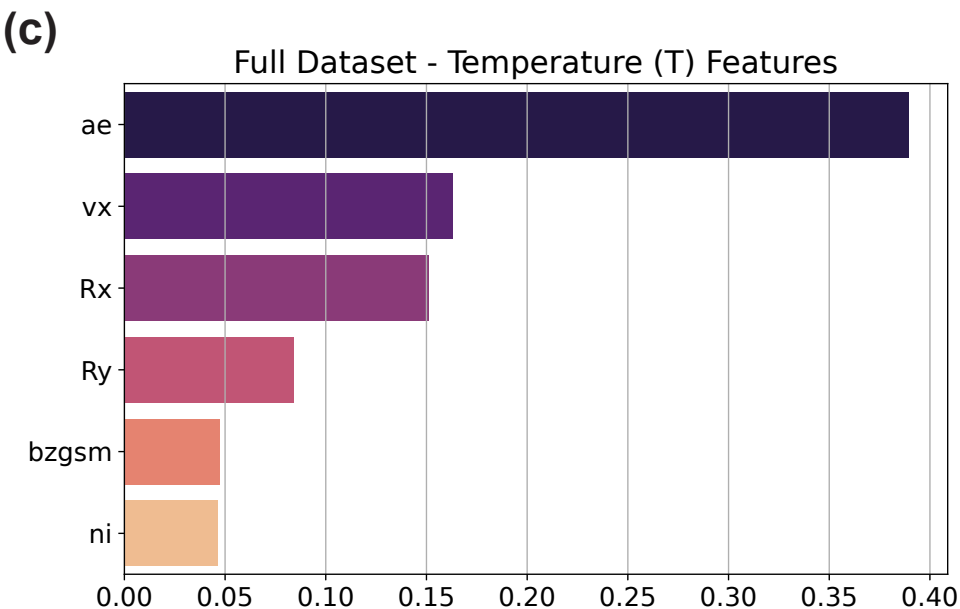


Quiet times

- More internally driven
- T is more external, maybe
- Density is largely dictated by location

Storms:

- More externally driven
- Location matters less

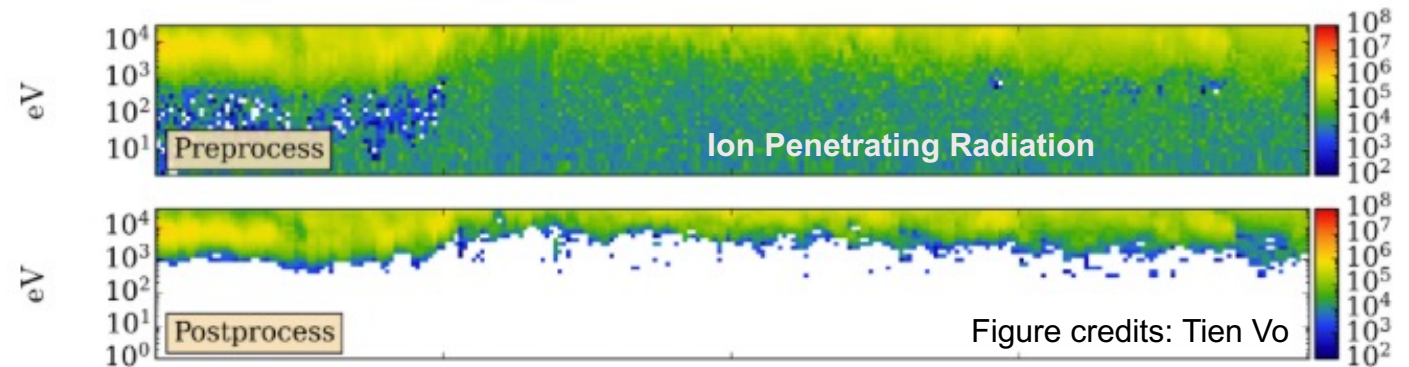
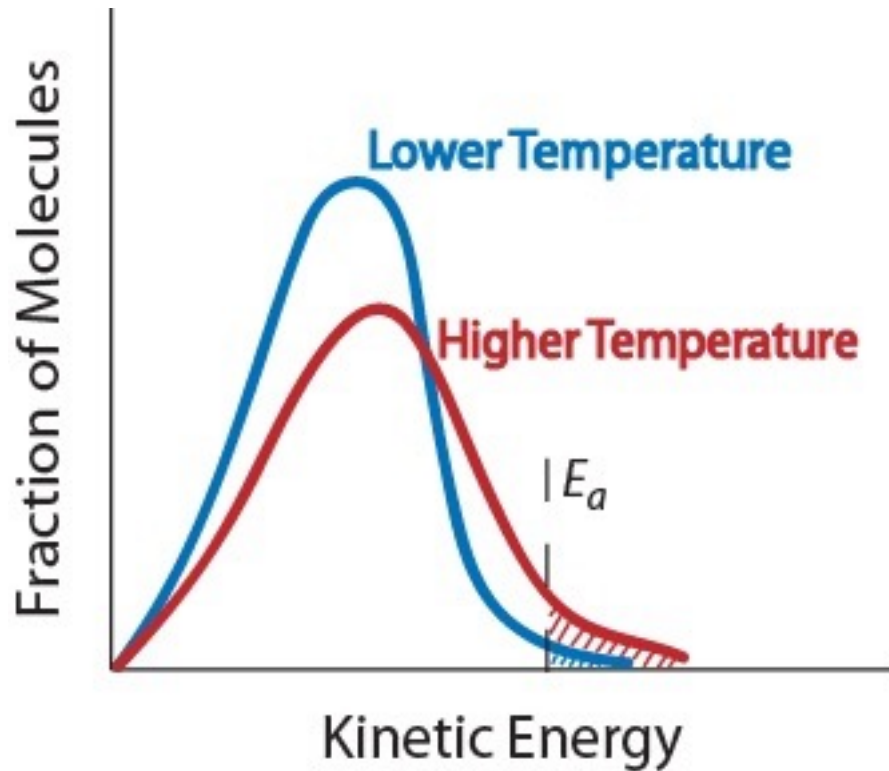


Issues with this analysis:

- Intercorrelation between variables is large (ae well correlated to vx)
- Same with SW types (dense solar wind is typically slower)
- Combines all time history (1h in this particular example)

Community Reminder on Temperature

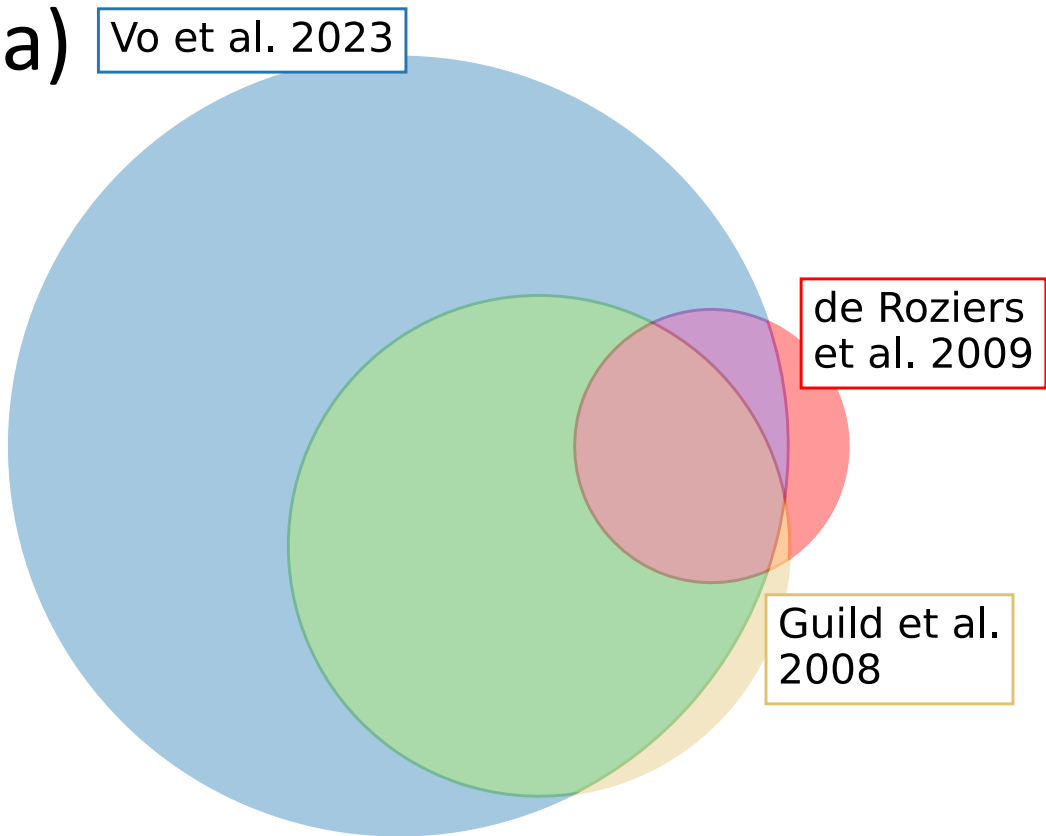
- Temperature is the 2nd plasma moment
- The higher the moment, the more uncertain because you rely more on the poorly sampled tails of the distribution.
- So, 0 and 1st moment (**Density and Velocity**) are usually ok, but Temperature, we got to be careful



$$T = \frac{m}{3k_B n} \int (\mathbf{v} - \mathbf{v}_b) \cdot (\mathbf{v} - \mathbf{v}_b) f(\mathbf{v}) d^3v$$

The temperature here was completely incorrect, and the velocity increased from about 200 km/s to over 1500 km/s.

Classifying plasma sheet is not trivial



Plasma Sheet Criteria			Number
Vo et al. 2023	de Roziers et al. 2009	Guild et al. 2008	
Yes	No	No	1,259,896
No	Yes	No	39,451
No	No	Yes	28,828
Yes	Yes	No	46,399
Yes	No	Yes	686,527
No	Yes	Yes	10,483
Yes	Yes	Yes	170,467

Note Vo+2023, had a multi-step process based on interval, this is just using the point-by-point classification

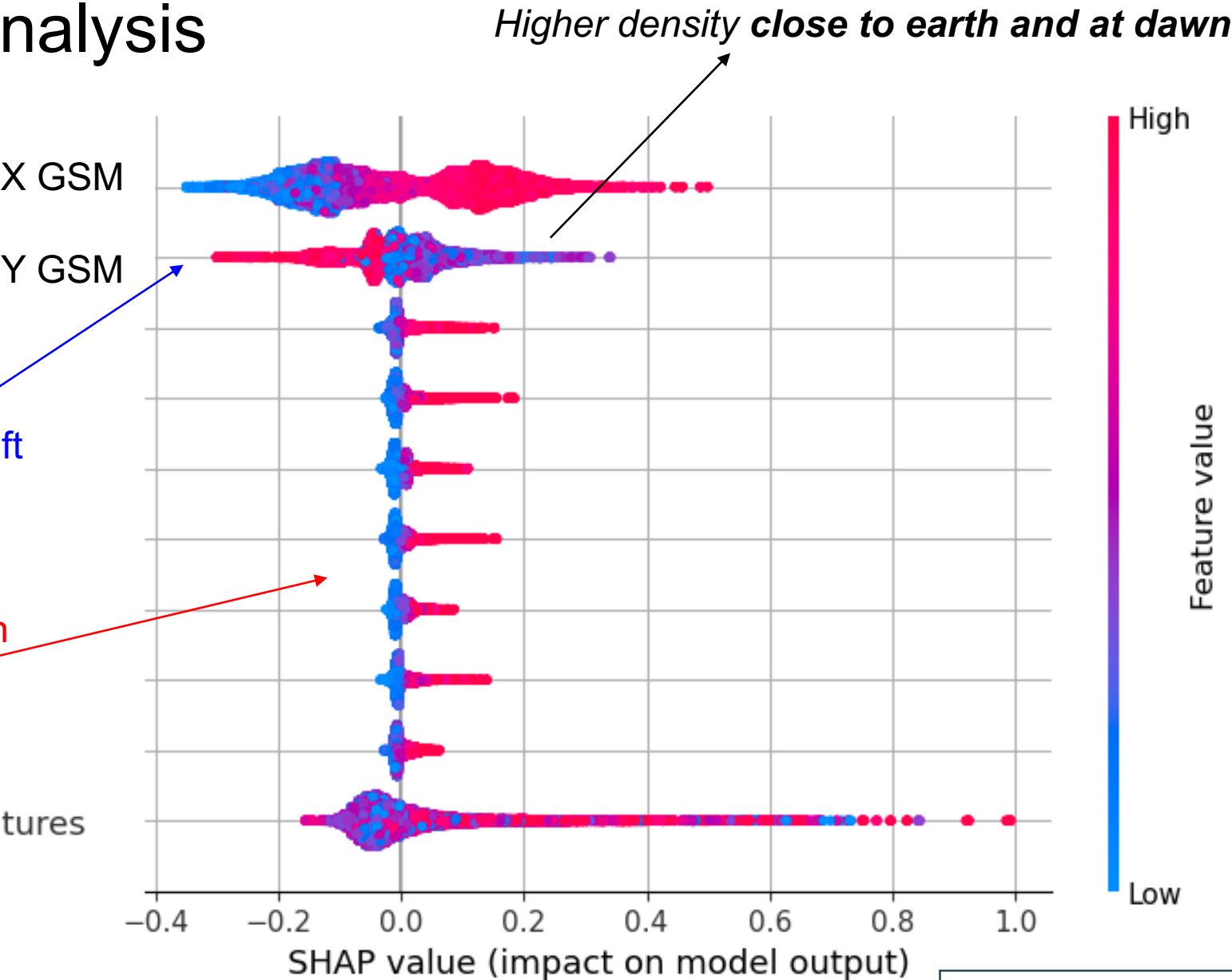
Feature Importance Analysis

Answer: In most cases (statistically):

Model is predominantly driven by spacecraft location

Solar wind input has lower effect, although cumulative history is still important

Sum of 85 other features



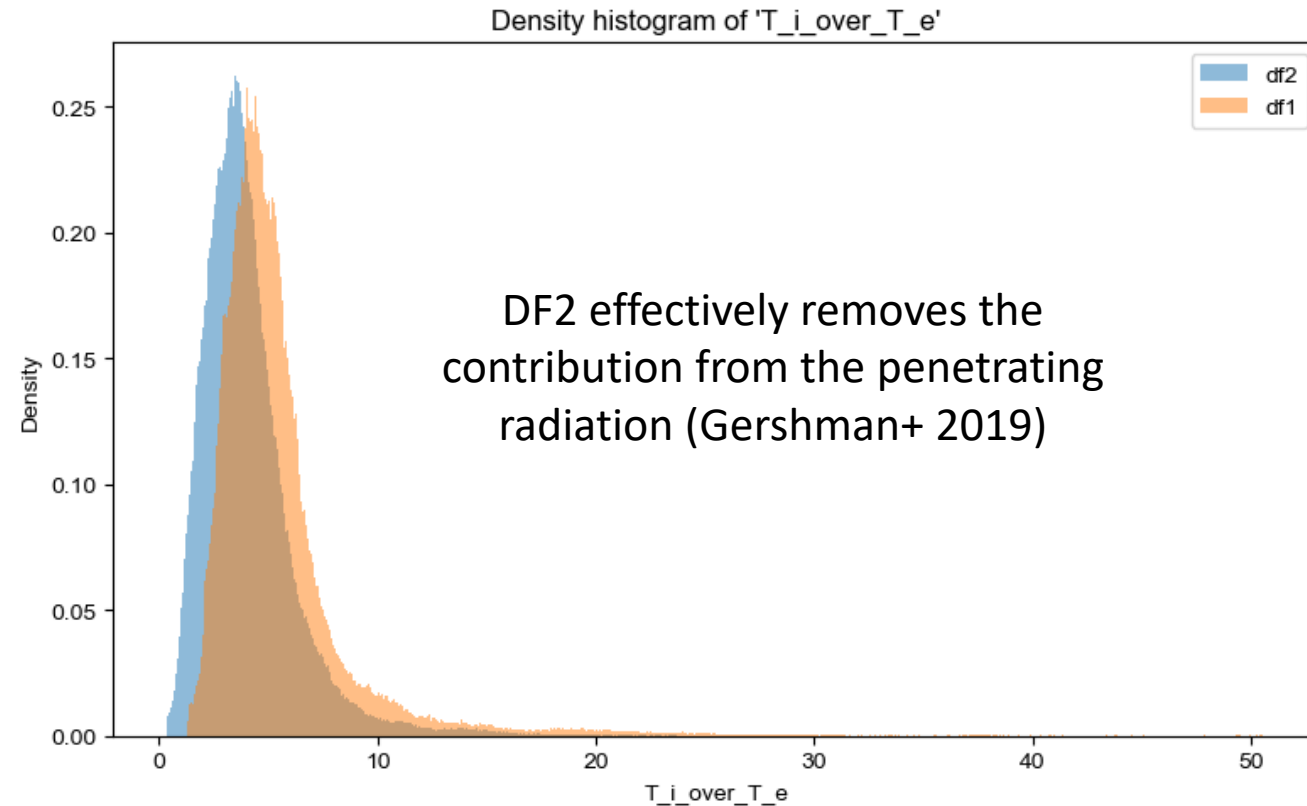
SHAP Values explain why a model made a specific prediction, by showing each feature's impact.

Blue/Red: Input value
Left/Right: Output value

MMS Ti/Te plasmashield ratio example (Full vs Partial moments)

mean df1 (full distribution moments): 5.5513

mean df2 (partial distribution moments): 4.0797



Key Message1: The mean differenced changed by 1.5 (>30%) simply by recalculating moments

Key Message2: A model with +30% is exciting, but we need to know if “*ground truth*” vary by the same magnitude